

Design note: the macro time-series inference tier

Status: approved design; implementation staged from v1.6.35.
Owner: Mico Mrkaic. Companion: DESIGN_VCE.md (the scores/meat interface this tier extends).

Scope and sequencing

Eight command families, implemented in dependency order:

1. `newey` — HAC (Bartlett/Newey-West) regression. FIRST because `pperron` and `lpirf` consume the long-run-variance machinery, which lands in `src/vce.c` as the promised “different M from the same s_i ”.
2. `dfuller` — Augmented Dickey-Fuller.
3. `pperron` — Phillips-Perron (reuses the HAC kernel).
4. `tsfilter` — `hp`, `bk`, `hamilton`.
5. `var` — reduced-form VAR by equation-wise OLS.
6. `vargranger` — Granger Wald tests from the last `var`.
7. `irf` — impulse responses from the last `var` (subset, below).
8. `lpirf` — local projections with Newey SEs.
9. `vecrank` — Johansen trace test. `vec` estimation is STAGED to a later release; `vecrank` answers the everyday question (“how many cointegrating relations?”) at a fraction of the surface.

Specification decisions

`newey`: `newey y xs [if] [in], lag(#)`. OLS point estimates; V from the `vce` module with Bartlett meat $M = \Omega_0 + \sum_{l=1..L} w_l (\Omega_l + \Omega_l')$, $w_l = 1 - l/(L+1)$, $\Omega_l = \sum_t s_t s_{t-l}'$. Small-sample factor $N/(N-k)$ (Stata’s). `t` stats with $N-k$ df. Requires `tsset`; gaps in time are an error (Stata behavior).

`dfuller`: `dfuller y [if] [in] [, lags(#) trend drift noconstant regress]`. Test regression Δy_t on y_{t-1} [, t] [, `const`] [, `lagged` Δ terms]. Reported: $Z(t) = \tau$ on y_{t-1} , the 1/5/10% critical values from the MacKinnon (2010) response surfaces (constant / trend / none variants), and the MacKinnon (1994) approximate p -value via the published regression coefficients. `regress` prints the underlying regression table. Critical values and p -values are tabulated approximations by construction; the Stata check pins agreement to displayed precision (4 decimals).

`pperron`: `pperron y [, lags(#) trend noconstant regress]`. DF regression WITHOUT lag augmentation; $Z(t)$ and $Z(\rho)$ corrected with

the Bartlett long-run variance of the residuals, default lag floor($4(T/100)^{(2/9)}$) (Stata's). Same MacKinnon furniture.

tsfilter: `tsfilter hp NEW = VAR [, smooth(#)]` with lambda default 1600 quarterly, 100 yearly, 129600 monthly (Stata's defaults by tsset frequency); solved exactly via the pentadiagonal system $(I + \lambda D'D)\tau = y$ with a banded LAPACK solve. `tsfilter bk NEW = VAR [, minperiod(#) maxperiod(#) k(#)]` Baxter-King symmetric truncated bandpass, defaults 6/32/12 quarterly scaled by frequency; the first and last k observations are missing. `tsfilter hamilton NEW = VAR [, h(#) p(#)]` Hamilton (2018) regression filter, defaults $h=8$, $p=4$ at quarterly (scaled by frequency); the residual is the cycle. All three create the NEW cycle variable; hp also honors `trend(NAME)`.

var: `var y1 y2 ... [if] [in] [, lags(1/p)]` default `lags(1/2)`. Equation-wise OLS with constant. Prints Stata's per-equation header block (Sample, obs, log-lik, AIC/HQIC/SBIC, per-equation R2/RMSE) and coefficient tables. Estimates stored for `vargranger/irf`: `A_1.. A_p`, Sigma (ML, divisor T), variable names, T.

vargranger: Wald χ^2 per (equation, excluded variable) pair on that variable's lags, plus the ALL row. Stata's table layout.

irf (subset): Stata's irf is a file-based subsystem (`irf create / irf set / active files`). tea keeps ONE active IRF set in memory, from the last `var: irf create [, step(#)]` default `step(8)`; computes `irf` and `oirf` `irf table [irf|oirf]` the response table, all pairs `irf graph [irf|oirf]` one combined SVG via the graph engine No .irf files, no multiple sets, no fevd this release; the point is the daily loop (`var -> irf table/graph`) not the archival system. Orthogonalized responses use the Cholesky factor of Sigma in the variable order given to `var` (Stata's convention).

lpirf: `lpirf y [if] [in] [, step(#) lags(#)]` Jorda local projections of y_{t+h} on the shock (y_t) and `lags(#)` controls of y , $h = 1..step$ (default 8), each horizon's SE Newey-West with lag h (Stata's default). Output: the coefficient/SE table per horizon. Multivariate `lpirf` (responses of other variables) is staged with `vec`.

vecrank: `vecrank y1 y2 ... [, lags(#) trend(constant|none)]` Johansen trace statistics from the canonical correlations of the VECM's concentrated system; 5% critical values from Osterwald-Lenum (1992) for the unrestricted-constant and no-constant cases. Prints the rank table with the selected rank starred, Stata-style.

Numerical ground rules

Everything is LAPACK/GSL, deterministic, and inside the 6-significant-digit display bound. Tabulated constants (MacKinnon response surfaces and p-value regressions, Osterwald-Lenum critical values) are embedded as static tables with their citations; they are data, not code, and COMPATIBILITY.md documents that p-values are approximations that may differ from Stata in the last displayed digit. `tools/stata_check_ts.do` mirrors every regression test so the IMF Stata run pins the whole tier.

Staged out (explicitly)

vec estimation, fevd, irf files/sets, multivariate lpirf, svar, xtunitroot, dfpls. Each becomes cheap once this tier's machinery exists; none blocks the daily loop this tier serves.